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# Application of Deep Learning to Detect Melanoma on Dermoscopy Images

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**Abstract.** Melanoma is a type of skin cancer that affects many people worldwide. Although it only comprises 1% of the total skin cancer incidence, 80% of deaths due to skin cancer is related to melanoma. Dermatologists are healthcare specialists that have a big role in the diagnosis of melanoma. However, these experts are not sufficient enough to cover a wide population. To augment this problem, creation of MelanomaENV2 is proposed. This research used the EfficientNet V2, and its variants, as template to create MelanomaENV2 model that can predict melanoma based on dermoscopy images. It is trained in 919 images: 400 for melanoma and 519 for nonmelanoma, which were sampled from the Human Against Machine dataset (HAM10000). Among the variants, efficientnetv2-l-21k-ft1k has the highest overall score among the evaluation metrics: 0.73 in precision, 0.69 in recall, 0.71 in F1 score, 0.88 in training accuracy, 0.72 in test accuracy, and 0.72 in Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

**Keywords.** deep learning, EfficientNet, melanoma, skin cancer, TensorFlow

## 1. Introduction

Melanoma has been known in human history since the 5th century B.C. It was a mysterious disease until modern medicine keeps on unraveling more of its causes and pathologic mechanism. Ultraviolet (UV) radiation is an important environmental factor for melanoma [1]. There is a heritable pattern for the spread of melanoma. Surgical removal is still the primary intervention to rid of melanoma [2]. The depth and spread of the melanoma correlates significantly to the prognosis of the disease. It is responsible for 80% of deaths due to skin cancer. Men have a slightly higher incidence rate of melanoma at 3.8 per 100,000 population than women having it at 3 per 100,000 population [3]. Through decades of research, there are discovered genes that are highly correlated to melanoma, namely: NRAS, BRAF, GNAQ, GNA-11, and CHEK2**110delC [2]. Prevention of melanoma is done through mass screening, use of sunscreen, and avoidance of UV sources. Vigilance on the screening and accuracy of diagnosis is lifesaving [2, 4, 5]. Because of the low number of dermatologist to population ratio, it raises the need for screening for melanoma via the use of deep learning models. Deep learning models for melanoma detection use different publicly available datasets containing dermoscopy images. Among the widely used datasets are HAM10000,

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BCN20000, ISIC, PH2, and ISBI. In terms of pre-trained models, the most popular are AlexNet, VGG, DenseNet, Inception, EfficientNet, ResNet, GoogleNet, and InceptionResNet.

Melanoma is a malignant lesion originating from melanocytes, with multiple invasive and non-invasive subtypes. Although melanoma accounts for only 1% of all skin cancer cases, it is responsible for 80% of related deaths. This means melanoma remains a significant global health concern. In 2020, an estimated 325,000 new cases were reported worldwide, with incidence rates higher in men (3.8 per 100,000) compared to women (3 per 100,000). Regional trends indicate a sharp increase in melanoma incidence from 1982 to 2011, such as in Australia (26.4 to 51.6 per 100,000) and the UK (5.8 to 19.8 per 100,000) [2]. Older men with thick melanoma lesions, lymph node involvement, or lesions on the trunk are at higher risk of fatal outcomes. While screening has advanced early detection, effectiveness varies significantly, with approximately 34,000 screenings needed to prevent one melanoma case [6]. Additionally, darker-skinned individuals often have melanoma occurrences in atypical areas like palms and soles [2].

## 2. Methodology

In this study, a novel deep learning model, MelanomaENV2, was developed using the EfficientNet V2 architecture to classify dermoscopy images as melanoma or nonmelanoma. EfficientNet V2 was chosen for its faster performance and reduced parameter size compared to its predecessors [7]. MelanomaENV2 leverages transfer learning by using a pre-trained ImageNet version for binary classification tasks. The dataset employed is HAM10000, which includes 10,000 dermoscopy images of both malignant and benign skin lesions, categorized into the following: melanocytic nevi, melanoma, benign keratosis, basal cell carcinoma, actinic keratosis, vascular lesions, and dermatofibromas [8].

The training data from HAM10000 was extracted to include 919 histopathologically confirmed images only. It was split into two classes: melanoma (400 cases) and nonmelanoma (519 cases). The dataset was partitioned into 80% training, which contains 736 dermoscopy images and 20% testing subsets, which contains 183 dermoscopy images, with preprocessing steps such as image rescaling to standardize 450 by 450 pixels inputs for compatibility with TensorFlow/Keras libraries. A sequential EfficientNet V2 architecture was implemented, incorporating a 20% dropout rate, sigmoid activation function, and binary cross-entropy loss. Training was conducted in Google Colab's free-tier environment with hardware optimized for GPU-based processing. Hardware configurations were: 12.7 Gb RAM, 78 Gb disk space, and 15 Gb GPU RAM. At the time of experiment and model creation, the version number is 2.11.0 for the TensorFlow GPU. Model evaluation utilized are precision, recall, F1-score, and AUC-ROC metrics. Visualization of training progress and predictions was achieved through Python libraries such as Matplotlib and Scikit-Learn, with all coding performed in a reproducible Google Colab notebook setup. Each of the EfficientNet V2 variant are tested to create MelanomaENV2: efficientnetv2-m, -l, -m-21k, -l-21k, -xl-21k, -m-21k-ft1k, -l-21k-ft1k, and -xl-21k-ft1k.

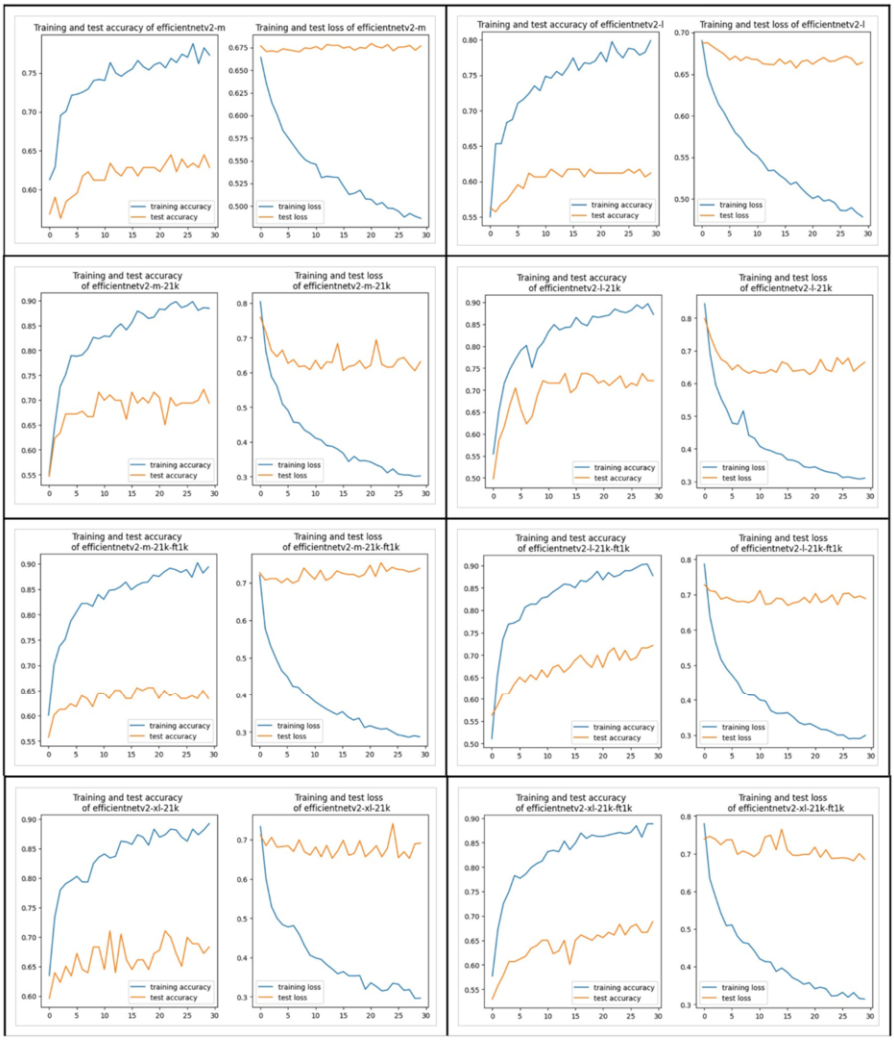


**Figure 1.** Summary plots of the accuracy and loss during the training and testing process for the different variants of EfficientNet V2.

## 3. Results

The different variants of EfficientNet V2 were trained on the same dataset for epochs. There are 117, 749, 410 parameters. However, only 2,562 are trainable parameters. Among the EfficientNet V2 variants shown in table 1; -l-21k-ft1k has the highest precision, -l-21k has the highest recall, -m-21k and -l-21k have the highest F1 score, -l-21k and -l-21k-ft1k have the highest test accuracy, and -l-21k-ft1k and -xl-21k-ft1k have the highest AUC-ROC score. Figure 1 demonstrates the summary of the graphical representation of the training and testing accuracy and loss during the 30-epoch duration.

**Table 1.** Summary evaluation of the performance metrics for each EfficientNet V2 variant.

| EfficientNet V2 variants | Precision | Recall | F1 score | Training accuracy | Test accuracy | AUC-ROC |
|---|---|---|---|---|---|---|
| -m | 0.56 | 0.77 | 0.65 | 0.77 | 0.62 | 0.67 |
| -l | 0.65 | 0.61 | 0.63 | 0.79 | 0.61 | 0.59 |
| -m-21k | 0.64 | 0.88 | 0.74 | 0.88 | 0.69 | 0.69 |
| -l-21k | 0.67 | 0.82 | 0.74 | 0.87 | 0.72 | 0.68 |
| -xl-21k | 0.41 | 0.7 | 0.52 | 0.89 | 0.68 | 0.62 |
| -m-21k-ft1k | 0.56 | 0.64 | 0.6 | 0.89 | 0.63 | 0.63 |
| -l-21k-ft1k | 0.73 | 0.69 | 0.71 | 0.88 | 0.72 | 0.72 |
| -xl-21k-ft1k | 0.71 | 0.67 | 0.69 | 0.89 | 0.69 | 0.72 |

## 4. Discussion

This research explores the application of EfficientNet V2 for melanoma classification using the HAM10000 dataset, focusing on binary classification tasks to distinguish melanoma from nonmelanoma skin lesions. Eight variants of EfficientNet V2 were tested, selected for their compatibility with the dataset's image resolution of 450 pixels. These included efficientnetv2-m, -l, -m-21k, -l-21k, -xl-21k, -m-21k-ft1k, -l-21k-ft1k, and -xl-21k-ft1k. Training and evaluation were conducted on Google Colab using GPU-based hardware, leveraging TensorFlow and Keras libraries to build and optimize models. Images were preprocessed through rescaling and partitioned into an 80/20 training-to-testing ratio, with the models trained for 30 epochs.

The implementation of EfficientNet V2 in developing the MelanomaENV2 model has demonstrated promising results in melanoma classification using dermoscopy images. Among the eight tested variants shown in figure 1, efficientnetv2-l-21k-ft1k achieved the highest performance in terms of accuracy, F1 score, and AUC-ROC, confirming the utility of deeper, pretrained models in medical image classification [7]. The model's moderate AUC of 0.72 suggests a reliable yet improvable level of discriminative ability, potentially useful in clinical triage systems.

Each model's performance was assessed using standard metrics, including accuracy, precision, recall, F1 score, and AUC-ROC. Efficientnetv2-l-21k-ft1k demonstrated superior performance in several metrics, including precision (0.73), test accuracy (0.72), and AUC-ROC (0.72), making it a strong candidate for deployment as the MelanomaENV2 model. In comparison, efficientnetv2-xl-21k exhibited the lowest precision (0.41), highlighting variability in model performance across different variants. The F1 score, a balance between precision and recall, ranged from 0.52 (efficientnetv2-xl-21k) to 0.74 (efficientnetv2-m-21k and -l-21k), with all models achieving acceptable performance. Despite the overall variability, efficientnetv2-l-21k-ft1k emerged as the most consistent and reliable variant for melanoma classification, underscoring its potential utility in clinical applications for skin lesion diagnosis.

## 3. Conclusion

One of the practical implications of this work lies in the potential application of MelanomaENV2 in primary care or community health settings. By providing an initial risk stratification for suspicious lesions, it could assist non-specialist clinicians or nurses in identifying high-risk cases for dermatologist referral [9, 10]. This is particularly important in regions with limited dermatological services, where delays in diagnosis can significantly affect prognosis [11].

However, the model's performance, while acceptable, remains below the threshold needed for autonomous clinical decision-making. Further improvements could be achieved through ensemble learning, metadata integration into the model itself, or training with a larger and more diverse dataset [12, 13]. The recommendations for future work include the incorporation of explainability features (e.g., Grad-CAM), testing in prospective clinical settings, and expansion into multi-class lesion classification to differentiate among various skin cancers [8, 14].

## References

[1] Scholz A. History of Skin Cancer. In P. Altmeyer, K. Hoffmann, & M. Stücker (Eds.), Skin Cancer and UV Radiation (pp. 20–27). Springer Berlin Heidelberg. 1997. https://doi.org/10.1007/978-3-642-60771-4_3

[2] Saginala K, Barsouk A, Aluru J, Rawla P, Barsouk A. Epidemiology of Melanoma. Medical Sciences. 2021;9(4), 63. https://doi.org/10.3390/medsci9040063

[3] Rebecca V, Sondak V, Smalley K. A brief history of melanoma: From mummies to mutations. Melanoma Research. 2012; 22(2), 114–122, https://doi.org/10.1097/CMR.0b013e328351fa4d

[4] Boniol M, Autier P, Boyle P, Gandini S. Cutaneous melanoma attributable to sunbed use: Systematic review and meta-analysis. BMJ (Clinical Research Ed.). 2012; 345, e4757. https://doi.org/10.1136/bmj.e4757

[5] Speckemeier C, Pahmeier K, Trocchi P, Schuldt K, Lax H, Nonnemacher M, Dröge P, Stang A, Wasem J, Neusser S. One-year follow-up healthcare costs of patients diagnosed with skin cancer in Germany: A claims data analysis. BMC Health Services Research. 2022;22(1), 771. https://doi.org/10.1186/s12913-022-08141-9

[6] Sacchetto L, Rosso S, Comber H, Bouchardy C, Broganelli P, Galceran J, Hackl M, Katalinic A, Louwman M, Robsahm T, Tryggvadottir L, Tumino R, Van Eycken E, Walsh P, Zadnik V, Zanetti R. Skin melanoma deaths within 1 or 3 years from diagnosis in Europe. International Journal of Cancer. 2021; 148(12), 2898–2905. https://doi.org/10.1002/ijc.33479

[7] Tan M, Le QV. EfficientNetV2: Smaller Models and Faster Training. ArXiv. 2021. https://doi.org/10.48550/arxiv.2104.00298

[8] Tschandl P. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions [Data set]. Harvard Dataverse. 2018. https://doi.org/10.7910/DVN/DBW86T

[9] Amer A. A Deep Learning Approach to Skin Cancer Detection in Dermoscopy Images. J Biomed Phys Eng. 2020;10(6):801–6. https://doi.org/10.31661/jbpe.v0i0.2004-1107

[10] Alwakid G, Gouda W, Humayun M, Sama NU. Melanoma Detection Using Deep Learning-Based Classifications. Healthcare (Basel). 2022;10(12):2481. https://doi.org/10.3390/healthcare10122481

[11] Alzubaidi L, Zhang J, Humaidi AJ, Al-Dujaili A, Duan Y, Al-Shamma O, et al. Review of deep learning: Concepts, CNN architectures, challenges, applications, future directions. J Big Data. 2021;8(1):53. https://doi.org/10.1186/s40537-021-00444-8

[12] Yang CH, Ren JH, Huang HC, Chuang LY, Chang PY. Deep Hybrid Convolutional Neural Network for Segmentation of Melanoma Skin Lesion. Comput Intell Neurosci. 2021;2021:9409508. https://doi.org/10.1155/2021/9409508

[13] Zhang H, Kalirai H, Acha-Sagredo A, Yang X, Zheng Y, Coupland SE. Piloting a Deep Learning Model for Predicting Nuclear BAP1 Immunohistochemical Expression of Uveal Melanoma from Hematoxylin-and-Eosin Sections. Transl Vis Sci Technol. 2020;9(2):50. https://doi.org/10.1167/tvst.9.2.50

[14] Xin C, Liu Z, Zhao K, et al. An improved transformer network for skin cancer classification. Comput Biol Med. 2022;149:105939. https://doi.org/10.1016/j.compbiomed.2022.105939